Design and Analysis of Spacecraft Systems

Spacecraft Mission Analysis Project

A low-cost small-satellite SAR Earth-Observation mission

WORKING DRAFT — fill highlighted fields before submission

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Individual mission parameters

Parameter	Value	Unit
Reference epoch	01 Aug 2026 00:00:00	UTC date
Equator cross time	10:30:00	UTC
Maximum latitude of ground track	55	deg
Initial orbit altitude	550	km
Initial orbit longitude (asc. node)	78 E	deg
Sandwich panel face-sheet thickness	0.50	mm
Honeycomb cell diameter	0.25	in
SAR swathe-mode operating power	120	W
SAR spot-mode operating power	200	W
SAR system mass	45	kg

1. Introduction

This report presents the mission and primary sub-system analysis for a low-cost small-satellite Earth-observation mission. The single payload is a synthetic-aperture radar (SAR) operating in a medium-resolution swathe mode and a high-resolution spot mode, tasked with monitoring vegetation and land use across tropical and temperate regions, with a secondary disaster-response role. The satellite flies a mid-inclination orbit to maximise dwell over the region of interest, with a six-year design life and the potential to grow into a constellation.

The analysis proceeds through five coupled parts: mission and orbit design (GMAT); the space environment and its design consequences (drag and MMOD); the electrical power budget; the mass budget, propulsion selection and launch vibration; and spacecraft configuration against the Rocket Lab Electron launch vehicle. Because the power and mass budgets depend on one another, and the vibration model depends on the closed mass, the final sections are iterated to a consistent solution.

2. Part 1 — Mission analysis

2.1 Method: deriving the Keplerian elements

The six classical orbital elements follow almost directly from the mission constraints. Two are set by geometry, three by the epoch conditions, and one (RAAN) is computed from sidereal time.

- Semi-major axis a: `a = R_E + h` with Earth radius R_E = 6378.14 km.
- Eccentricity e: near-circular, e ≈ 0. This simplifies eclipse, drag and coverage analysis and is appropriate for an EO SAR mission.
- Inclination i = maximum ground-track latitude. For a prograde orbit the sub-satellite latitude oscillates between ±i, so setting i equal to the max-latitude parameter satisfies both the coverage requirement and the mid-inclination requirement.
- True anomaly ν = 0 and argument of perigee ω = 0. The epoch condition is an ascending equator crossing (south to north), i.e. the argument of latitude is zero.
- RAAN Ω is computed so that the sub-satellite longitude at epoch equals the initial-longitude parameter: Ω `= GMST(epoch) + λ_0`, where GMST is the Greenwich Mean Sidereal Time at the reference-epoch UTC. The equator-crossing-time constraint fixes the epoch instant, which fixes GMST, which fixes Ω.

Illustrative result

Element	Symbol	Value (illustrative)
Semi-major axis	a	6928.14 km
Eccentricity	e	≈ 0 (circular)
Inclination	i	55.0 deg
RAAN	Omega	[compute from GMST + 78 deg]
Argument of perigee	omega	0 deg
True anomaly at epoch	nu	0 deg

NOTE: Replace RAAN with the value you compute from GMST at your epoch. State the GMST value and the conversion in your report so the marker can follow the working.

2.2 Propagator configuration (GMAT)

The force model required by the brief is configured in GMAT as follows: JGM-2 gravity field truncated to degree 4 and order 4; Jacchia-Roberts atmospheric density model; and the Sun and Moon included as point-mass third-body perturbations. The ground track is propagated for ten days from the reference epoch and exported. The complete GMAT script is provided in Appendix A.

2.3 Eclipse fraction (Part 1b)

For a near-circular orbit the fraction of the period spent in eclipse is obtained from the cylindrical-shadow model:

$$f_{ecl} = \frac{1}{180} \, arccos\left(\frac{\sqrt{h^2 + 2R_E h}}{(R_E + h)\cos\beta}\right) \quad [deg]$$

where beta is the beta angle between the orbit plane and the Sun line. The maximum eclipse occurs at beta = 0. The eclipse fraction should also be read directly from the GMAT EclipseLocator and cross-checked against this formula; the two should agree to within a couple of percent.

Illustrative result (beta = 0, worst case)

```
sqrt(550^2 + 2*6378.14*550) = 2705.3 km;  ratio = 2705.3 / 6928.14 = 0.3905;  arccos
= 67.0 deg.
```

Eclipse fraction = 67.0 / 180 = 0.372 (37.2% of the period). With T = 95.7 min, eclipse ≈ 35.6 min and sunlight ≈ 60.1 min.

NOTE: These sunlit and eclipse durations feed the power budget (Part 3) directly. Keep them consistent.

2.4 Ground-station selection (Part 1c)

The requirement is an average of 1.5 hours of ground contact per day for full data transfer. The procedure is:

- In GMAT, define each candidate station as a GroundStation and create a ContactLocator with a chosen minimum elevation mask (5 to 10 deg is typical).
- Propagate for a representative period (at least the 10-day window) and extract per-pass access durations for each station.
- Sum the daily access per station, then select the smallest set of stations whose combined daily average meets or exceeds 1.5 hours.

Physical reasoning to include: for a 55 deg inclination the ground track does not reach high latitudes, so a station such as Kiruna (67.85 N) yields short, low-elevation passes and is inefficient. Stations within or near the inclination band, spread in longitude to catch successive ascending and descending passes, give the best return. State your final selection and the daily-contact total in a table.

NOTE: Populate a station table with your GMAT-derived daily contact minutes and mark which stations you selected and why.

3. Part 2 — Space environment

3.1 Environment features and design consequences (Part 2a)

At the selected low-Earth altitude the dominant environmental drivers, and their principal design consequences, are:

- **Neutral atmosphere and drag.** Residual density decelerates the satellite, lowering the orbit. Consequence: a propulsion system and propellant budget for orbit maintenance, plus disturbance torques the attitude system must manage.
- **Atomic oxygen.** Highly reactive AO erodes polymers and unprotected surfaces. Consequence: AO-resistant coatings and material selection for exposed surfaces, blankets and array substrates.
- **Ionising radiation.** Trapped protons and electrons (including the South Atlantic Anomaly) and galactic cosmic rays. Consequence: total-ionising-dose rating of parts, single-event-effect mitigation, and degradation allowance for solar cells (see Part 3).
- **Solar UV and thermal cycling.** Repeated eclipse-to-sunlight transitions drive large thermal swings. Consequence: a thermal-control subsystem, coatings with stable optical properties, and fatigue-tolerant structural joints.
- **Plasma and surface charging.** Differential charging risks electrostatic discharge. Consequence: conductive surface treatments and a single-point grounding scheme.
- **Micrometeoroid and orbital debris (MMOD).** Hypervelocity impacts can penetrate the skin. Consequence: a shielding assessment (Part 2c) and, if needed, sandwich-panel sizing.
- **Geomagnetic field.** Interaction with residual magnetic dipole produces torque. Consequence: magnetic cleanliness and/or magnetorquers for momentum management.

3.2 Drag force and orbit-maintenance delta-v (Part 2b)

The drag force on the satellite is:

$$F_{drag} = \frac{1}{2} \rho V^2 C_d A$$

with drag coefficient C_d ≈ 2.2 (typical free-molecular value), orbital velocity V = sqrt(mu / (R_E + h)), ram cross-sectional area A (include the deployed Earth-facing SAR antennas), and atmospheric density rho taken from the online NRLMSIS model at your altitude, epoch and solar conditions.

The delta-v that must be supplied over the mission to replace the momentum lost to drag is approximated by:

$$\Delta v_{drag} \approx \frac{F_{drag}}{m} \cdot t_{mission}$$

A margin is then applied to cover the large uncertainty in atmospheric density (which can vary by nearly an order of magnitude across the solar cycle) and in C_d and A at this design stage. A factor of two on the drag delta-v is defensible and should be stated explicitly.

Illustrative result

```
V = sqrt(398600.44 / 6928.14) = 7.585 km/s;  rho = 1.0e-13 kg/m^3 [from NRLMSIS];  A = 1.5 m^2;  m = 120 kg.

F_drag = 0.5 * 1.0e-13 * 7585^2 * 2.2 * 1.5 = 9.5e-6 N.

delta_v_drag = (9.5e-6 / 120) * (6 * 3.156e7 s) = 15.0 m/s;  with x2 margin = 30 m/s.
```

NOTE: The margined drag delta-v (30 m/s illustrative) carries into the propulsion sizing in Part 4a. Use your own NRLMSIS density and your own ram area.

3.3 MMOD penetration assessment (Part 2c)

3.3.1 Critical impact diameter — Whipple shield (Part 2c.i)

The aluminium honeycomb sandwich panel is modelled as a Whipple shield: the front face sheet acts as the bumper, the rear face sheet as the rear wall, and the standoff is taken as twice the honeycomb cell diameter per Section 6.1 of the MMOD handbook [1]. At a normal-incidence impact velocity of 10.5 km/s the impact is in the hypervelocity regime (V*cos(theta) >= 7 km/s for aluminium), so the New Non-Optimum ballistic-limit equation for critical projectile diameter applies [1]:

$$d_c = 3.918 \cdot \frac{{t_w}^{2/3} S^{1/3} \left(\frac{\sigma}{70}\right)^{1/3}}{{\rho_p}^{1/3} {\rho_b}^{1/9} (V \cos\theta)^{2/3}}$$

with rear-wall thickness t_w and standoff S in cm; rear-wall yield stress sigma in ksi (Al2024-T81 ≈ 65 ksi); projectile and bumper densities rho_p ≈ rho_b ≈ 2.7 g/cm^3 (standard aluminium-debris assumption); V in km/s; theta measured from the surface normal.

Illustrative result

```
t_w = 0.05 cm;  S = 2 * 0.25 in = 1.27 cm;  sigma = 65 ksi;  V*cos(theta) = 10.5 km/s.
```

```
d_c = 3.918 * (0.05)^0.667 * (1.27)^0.333 * (65/70)^0.333 / [ 2.7^0.333 * 2.7^0.111 * 10.5^0.667 ]
```

```
d_c = 0.5619 / 7.457 = 0.0754 cm ≈ 0.75 mm.
```

NOTE: Confirm the Al2024-T81 yield stress used and the exact hypervelocity-regime bound from the handbook [1] in your report.

3.3.2 Probability of a penetrating impact (Part 2c.ii)

Using the critical diameter, the debris flux for a projectile of that size or larger is read from the flux curves in Appendix A of the brief, interpolating linearly in diameter and, where the orbit altitude falls between tabulated altitudes, in altitude as well. The expected number of penetrating impacts over the mission and the probability of at least one impact (Poisson statistics) are:

$$N = \Phi \cdot A_{exp} \cdot t_{mission} \qquad P(\geq 1) = 1 - e^{-N}$$

Illustrative result

```
d_c = 0.000754 m at 550 km. Interpolating between 0.000692 m (0.01309) and 0.000832 m (0.004838):
```

```
Phi = 0.01309 + 0.443*(0.004838 - 0.01309) = 0.00944 /m^2/yr.
```

```
N = 0.00944 * 6 m^2 * 6 yr = 0.34;  P(>=1) = 1 - exp(-0.34) = 0.29 (about a 29% chance over the 6-year life).
```

NOTE: Use your own critical diameter, your spacecraft exposed area, and interpolate at your orbit altitude (interpolate between the 500/550/600 km columns if your altitude is not tabulated). Discuss whether the resulting probability is acceptable and, if not, how shielding could be increased.

4. Part 3 — Power budget

The power budget is built from a mode-by-mode load list (Tables 3.1 of the brief plus the spreadsheet payload powers), split across the sunlit and eclipse portions of the orbit, applying the stated duty cycles: spot mode 10% of the time over both the day and night faces; swathe mode 40% of the time over the day face only; and the propulsion system never running while the SAR is active.

4.1 Load list and averaged power

Construct a table of each subsystem's power in each condition, then compute the orbit-averaged demand separately for the sunlit and eclipse phases. A schematic form (fill with your subsystem powers):

Load condition	Contributors	Avg power (illustrative)
Sunlit average (P_d)	C&DH + SAR duty mix + comms + power + ACS	130 W
Eclipse average (P_e)	C&DH + SAR spot duty + battery heater + power + ACS	90 W

NOTE: Replace with your computed averages built from Table 3.1 and your spreadsheet SAR powers, weighted by the duty cycles and the sunlit/eclipse split from Part 1b.

4.2 Battery capacity (Part 3a)

The battery must supply the eclipse energy within the maximum routine depth of discharge, allowing for end-of-life capacity fade:

$$C_{req} = \frac{P_e \, T_e}{DoD \cdot f_{EOL}}$$

with maximum DoD = 40% and end-of-life capacity factor f_EOL = 70% (Table 3.2). Battery mass follows from the 100 Wh/kg energy density.

Illustrative result

```
E_e = 90 W * 0.593 h = 53.4 Wh;  C_req = 53.4 / (0.40 * 0.70) = 191 Wh;  mass =
191/100 = 1.9 kg.
```

4.3 Solar array area (Part 3b)

The array must carry the sunlit loads and recharge the battery within the sunlit period, accounting for transmission and charging efficiencies (both 90%, Table 3.2):

$$P_{sa} = \frac{P_d}{\eta_t} + \frac{E_e \,/\, \eta_c}{\eta_t \, T_d}$$

$$A_{array} = \frac{P_{sa}}{S \cdot \eta_{cell,EOL} \cdot \eta_{pack}}$$

with solar constant S = 1361 W/m^2, packing efficiency 90% (Table 3.2), and cell efficiency taken from Table 3.3 degraded to the 1-MeV fluence implied by the orbit and six-year life (end-of-life value). Choose the area-density (0.25 kg/m^2 body-mounted or 1.1 kg/m^2 rigid panel, Table 3.2) for the array mass that feeds Part 4.

Illustrative result

```
P_sa = 130/0.9 + (53.4/0.9)/(0.9*1.001) = 144.4 + 65.8 = 210 W.
```

```
A_array = 210 / (1361 * 0.287 * 0.90) = 0.60 m^2.  Rigid-panel mass = 1.1 * 0.60 =
0.66 kg.
```

NOTE: If you select the 200 W Hall thruster in Part 4a, add its power demand (when active, outside SAR operation) to the array/battery sizing and re-close the budget. This is the Part 3 to Part 4 coupling.

5. Part 4 — Spacecraft mass and vibration

5.1 Propulsion selection and propellant mass (Part 4a)

The propulsion system must provide 18 m/s (orbit acquisition, collision avoidance, manoeuvring) + 2 m/s (losses and end-of-life residual) + the margined orbit-maintenance delta-v from Part 2b. The propellant mass follows from the rocket equation:

$$m_{prop} = m_{dry}\left(e^{\frac{\Delta v}{I_{sp}\, g_0}} - 1\right) \quad (g_0 = 9.80665\, m/s^{2})$$

Both options are evaluated and traded. The trade is genuine and can flip depending on your total delta-v: cold gas carries a heavy 3x-propellant tank factor but needs no power, while the Hall thruster is very propellant-efficient but adds 200 W of demand (larger array and battery) and more dry mass.

Illustrative comparison (total delta-v = 50 m/s, m_dry ≈ 120 kg)

Quantity	Cold gas (N2)	Xenon Hall
Isp	72 s	1660 s
Propellant mass	8.8 kg	0.37 kg
Tank mass	26.4 kg (3x)	0.55 kg (1.5x)
System dry mass	0.25 kg	3.5 kg
Operating power	0 W	200 W
Total propulsion mass	≈ 35.5 kg	≈ 4.4 kg

Discussion: at this low total delta-v the cold-gas propellant itself is modest, but the 3x tank factor makes the cold-gas system heavy. The Hall option is far lighter on mass but imposes a 200 W power penalty and greater complexity, and its 9 mN thrust implies long burn durations. Select one, justify it against the mission's cost, mass, power and complexity priorities, and carry the choice consistently into the mass and power budgets.

NOTE: The winner depends on YOUR delta-v (which depends on your drag) and YOUR closed mass. Recompute both columns with your numbers before committing, and state the decision rationale explicitly.

5.2 Mass budget (Part 4b)

The mass budget is assembled per Table 4.2, with structure at 20% of dry mass and the power-system mass equal to 16 kg plus the battery and array masses from Part 3. Because structure, propellant and power mass all depend on the total, the budget is iterated to convergence and a margin is carried.

Subsystem	Basis	Mass (illustrative)
Payload (SAR)	Spreadsheet	45.0 kg
Structure	20% of dry mass	20.0 kg
ADCS	Fixed	12.0 kg
Propulsion (dry)	Part 4a	0.25 kg
Propellant + tank	Part 4a	35.2 kg

Subsystem	Basis	Mass (illustrative)
Power system	16 + battery + array	18.6 kg
Thermal control	Fixed	4.4 kg
Wiring and harness	Fixed	4.5 kg
Communications	Fixed	4.1 kg
C&DH	Fixed	7.9 kg
TOTAL (with margin)	Iterated	≈ 120 kg

NOTE: These figures are illustrative and internally approximate only. Rebuild the whole column from your parameters and iterate until structure = 20% of the final dry mass is self-consistent. Carry a stated system margin (e.g. 10-20%).

5.3 Launch vibration — minimum stiffness (Part 4c)

The primary axial frequency must exceed 30 Hz and secondary structures 40 Hz to avoid coupling with the Electron dynamic environment. The lumped-mass model of Figure 4.1 is a six-degree-of-freedom system in the coordinates x1..x6. The natural frequencies are the roots of the eigenvalue problem:

$$det(K - \omega^2 M) = 0, \quad f = \frac{\omega}{2\pi}$$

The mass matrix is diagonal in the six coordinates (x1 = m1 left, x2 = m1 right, x3 = m3, x4 = m2 upper, x5 = m4, x6 = m2 lower):

$$M = diag(m_1, m_1, m_3, m_2, m_4, m_2)$$

with the mass definitions from Table 4.3: m1 = 2.7 kg (SAR antenna); m2 = 0.5*(M_s - 2*m1 - m3 - m4) where M_s is the total satellite mass from Part 4b; m3 = payload-electronics mass (spreadsheet, corrected for the separate SAR antennas); m4 = total fuel-plus-tank mass from Part 4a; and k3 = 0.5*k1.

The stiffness matrix is assembled from the spring connectivity: each spring k connecting coordinates i and j adds +k to K(i,i) and K(j,j) and -k to K(i,j) and K(j,i); a spring to ground adds +k to the diagonal only. Read the exact connections from Figure 4.1 and populate K accordingly, then impose f1 (the fundamental axial mode) > 30 Hz and solve for the minimum k1 and k2.

NOTE: The exact spring topology must be taken from Figure 4.1 in your brief; assemble K from it using the rule above rather than assuming. Verify against the figure before trusting any stiffness value.

Illustrative single-mode check

As a sanity bound, the top mass m3 on spring k1 gives a single-DOF frequency f = (1/2pi)*sqrt(k1/m3). Requiring f > 30 Hz sets a lower bound on k1:

$$k_1 > (2\pi \cdot 30)^2 \cdot m_3$$

```
e.g.  m3 = 8 kg   ->   k1 > 2.84e5 N/m
```

Use the full 6-DOF eigen-solution for the reported values; the single-DOF bound is only a check that the fundamental is in the right range.

6. Part 5 — Spacecraft configuration

6.1 Suitability of the Electron launch vehicle (Part 5a)

The Electron is assessed against the closed spacecraft mass and volume using the Payload User's Guide [2]. The assessment should confirm each of the following against the guide and state the resulting design constraints:

- Mass capability to the target orbit versus the closed satellite mass, with positive margin.
- Dynamic (usable) volume of the fairing versus the stowed spacecraft, including the folded, body-latched SAR antennas.
- Mechanical interface and separation system (mass, ring diameter, separation shock).
- Sine, random and shock environments, which set the 30 Hz primary / 40 Hz secondary minimum-frequency requirements used in Part 4c.
- Centre-of-gravity limits, cleanliness, and RF/telemetry constraints.

NOTE: Insert the exact figures (payload-to-orbit for your altitude and inclination, fairing usable envelope, separation-system options, and the coupled-loads levels) from Electron Payload User's Guide v7.0 on Brightspace, and cite them. Do not rely on remembered numbers for these.

Conclude with a clear statement on suitability: whether Electron closes the mission with margin, and what the binding constraint is (typically volume or minimum frequency for a small dense SAR bus).

6.2 Configuration sketch (Part 5b)

Provide a labelled sketch (not dimensioned) showing: the two SAR antennas folded and latched to the body sides for launch and deployed on the Earth-facing side on orbit; the solar array (body-mounted or deployable per your Part 3 choice); the propulsion module and thruster location; the communications antenna oriented to the ground stations; and the attitude sensors and actuators. Indicate the Earth-facing, ram and anti-ram directions.

NOTE: Insert your sketch here (hand drawing or CAD screenshot). Ensure the payload, arrays and each primary subsystem are located and labelled, as required by the brief.

7. Conclusions

[Write once numbers are final.] State the closed orbit and its coverage performance; the eclipse fraction and ground-station solution; the drag and MMOD findings with the maintenance delta-v and penetration probability; the closed and consistent power and mass budgets; the propulsion selection with justification; the minimum stiffnesses satisfying the launch-frequency requirements; and the overall verdict on Electron suitability and mission feasibility. Note the principal uncertainties (atmospheric density, mass margin) and how they were bounded.

References

[1] E. L. Christiansen, Handbook for Designing MMOD Protection, NASA JSC-64399 Version A, January 2009.

[2] Rocket Lab USA Inc., Electron Payload User's Guide, Version 7.0, November 2022.

[3] NRLMSIS atmosphere model, NASA CCMC Instant-Run: https://kauai.ccmc.gsfc.nasa.gov/instantrun/nrlmsis/

[4] Add course notes / SMAD (Space Mission Analysis and Design) references as used.

Appendix A — GMAT script

Template script for Part 1. Replace the highlighted epoch, element and station values with your own, run, and export the ground-track and contact reports. Verify the force-model settings match the brief (JGM-2 4x4, Jacchia-Roberts, Sun and Moon point masses).

```
%--- Spacecraft ---
Create Spacecraft SAT;
SAT.DateFormat = UTCGregorian;
SAT.Epoch = '01 Aug 2026 10:30:00.000';   % <-- your epoch @ equator crossing
SAT.CoordinateSystem = EarthMJ2000Eq;
SAT.DisplayStateType = Keplerian;
SAT.SMA  = 6928.14;      % <-- R_E + your altitude (km)
SAT.ECC  = 0.0001;       % near-circular
SAT.INC  = 55.0;         % <-- your max ground-track latitude (deg)
SAT.RAAN = 0.0;          % <-- compute: GMST(epoch) + initial longitude
SAT.AOP  = 0.0;
SAT.TA   = 0.0;          % ascending node at epoch

%--- Force model: JGM-2 (4x4), Jacchia-Roberts, Sun+Moon ---
Create ForceModel FM;
FM.CentralBody = Earth;
FM.PrimaryBodies = {Earth};
FM.GravityField.Earth.Degree = 4;
FM.GravityField.Earth.Order  = 4;
FM.GravityField.Earth.PotentialFile = 'JGM2.cof';
FM.Drag.AtmosphereModel = JacchiaRoberts;
FM.PointMasses = {Sun, Luna};
FM.SRP = Off;

Create Propagator PROP;
PROP.FM = FM;

%--- Ground stations (add the set you select) ---
Create GroundStation Kourou;
Kourou.CentralBody = Earth;
Kourou.StateType = Spherical;
Kourou.HorizonReference = Ellipsoid;
Kourou.Location1 = 5.2;      % lat (deg)
Kourou.Location2 = -52.7;    % lon (deg)
Kourou.Location3 = 0.005;    % alt (km)

Create ContactLocator CL;
CL.Target = SAT;
CL.Observers = {Kourou};
CL.Filename = 'Contact.txt';

Create EclipseLocator EL;
EL.Spacecraft = SAT;
EL.Filename = 'Eclipse.txt';

Create OrbitView GT;   % configure a GroundTrackPlot in the GUI as well

%--- Mission sequence: propagate 10 days ---
BeginMissionSequence;
Propagate PROP(SAT) {SAT.ElapsedDays = 10};
```

Appendix B — NRLMSIS density run

Record the inputs and the returned density here so the drag calculation is traceable:

- Model: NRLMSIS (CCMC Instant-Run).
- Date / time: [your reference epoch].
- Altitude: [your altitude, km]; latitude/longitude representative of the orbit.
- Solar and geomagnetic indices used (F10.7, Ap): [state values / assumption].
- Returned total mass density rho: [value, kg/m^3] — used in Part 2b.

NOTE: Paste a screenshot or the tabulated output of your NRLMSIS run for the report appendix.